# School of Electrical Engineering and Computer Sciences
# National University of Sciences and Technology

## Faulty of Computing (FOC)
## Industry Advisory Board Meeting Minutes
## Held on 02nd May 2024

**In attendance**

| S.No. | Name | Designation/Organization/Company |
|---|---|---|
| 1. | Dr. Faisal Shafait (In-Chair) | Professor, Associate Dean FOC, NUST-SEECS |
| 2. | Dr. Rafia Mumtaz | Professor, Head SE Department, NUST-SEECS |
| 3. | Dr. Imran Malik | Associate Professor, Head CS Department, NUST-SEECS |
| 4. | Dr. Muhammad Bilal Ali | Assistant Professor, UG Coordinator, NUST-SEECS |
| 5. | Dr. Hashir Moheed Kiani | Assistant Professor, FYP Coordinator, NUST-SEECS |
| 6. | Dr. Mehdi Hussain | Associate Professor, NUST-SEECS |
| 7. | Mr. Taufique ur Rehman | RVF, NUST-SEECS |
| 8. | Ms. Zunera Zahid | Lecturer, NUST-SEECS |
| 9. | Ms. Mehvish Kiran | Lecturer, DLQEC, NUST-SEECS |
| 10. | Mr. Umer Nasim | Software Engineer/ Department Coordinator, NUST-SEECS |
| 11. | Ms. Samine Ikram | Vice Principal, Crescent Model School, Lahore |
| 12. | Ms. Saba Kamran | Digital Marketing Executive, Leverify |
| 13. | Brig. Dr. Abdul Razzaq | HoD, NUST-MCS |
| 14. | Dr. Wakeel Mahmood | Director, PAF |
| 15. | Mr. Khayyam Hussain | HoD, Roots |
| 16. | Mr. Rafay Ahmed Khan | ML Lead, Infinite Horizons |
| 17. | Mr. Muhammad Mubeen | AI Lead, Melitor ITS |
| 18. | Mr. Abdul Rehman | ML Engineer, Red Marker System |
| 19. | Mr. Fuqan Aziz | Sr. ML Engineer, Dcube Tech |
| 20. | Dr. Hammad A. Quresshi | Chief Tech Officer, Panni Management |
| 21. | Mr. Khawaja Sarmad Jawad | AM Talent Acquisition, Sotoo Pvt. Ltd. |
| 22. | Mr. Syed Juniad Ahmed | COO, Sotoo Pvt. Ltd. |
| 23. | Mr. Saad Liaqat | CTO, Sotoo Pvt. Ltd. |

**Venue: CIPS Board Room, Meeting time: 1400 hrs – 1600 hrs**

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**Agenda Points / Discussion**

- Dr Faisal greeted all the participants and provided an overview of the agenda items and the degree programs, the vision and mission of the department and how the curriculum is designed to achieve this vision and mission.
- The Industry Advisory Board (IAB) meeting comprises industry representatives from prestigious companies and faculty members from SEECS. The IAB meetings aim to enhance the academia-industry linkage and its impact through industrial feedback on the design, development, delivery, and evolution of the curriculum of computing degree programs.
- The industry representatives briefed the house about their companies and the different products and technologies on which they are working.
- The meeting started with a discussion on the following points.
    - SEECS At A Glance
    - Washington Accord – PEC Accreditation of Software Engineering Department
    - BSCS Degree Program – NCEAC Accreditation
    - Key Highlights Since the Last IAB Meeting
    - Student Exposure
    - Pressing Issues Requiring IAB's Attention
- Dr Hashir presented that during the Washington Accord (WA) PEC Accreditation Visit on October 30, 2023, the BE Software Engineering Program at SEECS was assessed. We are delighted to announce that our BESE program has successfully obtained accreditation for a duration of three years, applicable to batches up to 2022. This accreditation highlights the commitment to academic excellence and rigor maintained by the SE department.
- The NCEAC Accreditation visit took place on December 27th, 2023. Accreditation was granted for the BSCS intake years of 2020, 2021, and 2022. The department and faculty were commended for their efforts in achieving a competitive advantage over other universities. A significant focus was placed on identifying opportunities for improvement in OBE and streamlining ongoing efforts. Previously, NCEAC conducted a change of scope visit aimed at expanding the intake of the BSCS degree program. NCEAC's change of scope visit appreciated resources at hand and allowed SEECS to expand its BSCS Program
- Periodic Continuous Quality Improvement (CQI) meetings are convened with the aim of enhancing the curriculum. During recent sessions, the department has reviewed and updated the Course Learning Outcomes (CLOs) and Bloom's Taxonomy (BT) levels for multiple courses. Furthermore, additional CLOs have been incorporated to encompass the psychomotor and affective domains, ensuring a balanced mapping of Program Learning Outcomes (PLOs). In alignment with these updates, laboratory rubrics have

been adjusted to effectively evaluate the psychomotor and affective domains. Formal training sessions were conducted for Learning Engineers (LEs) and Faculty Members (FMs) to enhance their understanding of Outcome-Based Education (OBE).

- The following courses are included during the last two years

| Title | CHs | Title | CHs |
|---|---|---|---|
| Occupational Health and Safety | 1+0 | Big Data Analytics | 2+1 |
| Advance Topics in Computing (Large Language Models) | 3+0 | Deep Learning | 3+1 |
| Cloud Computing | 2+1 | Agent Based Software Engineering | 2+1 |
| Information Security | 3+0 | Visual Programming | 2+1 |
| Internet of Things | 3+0 | Data Encryption and Security | 2+1 |
| Embedded Systems | 2+1 | Augmented and Virtual Reality | 3+1 |
| | | Reinforcement Learning | 2+1 |

- During the past year, the Faculty of Computing (FOC) directed its attention towards prioritizing stream-based and hands-on electives within the domains of Artificial Intelligence, Cyber Security, and the Internet of Things.
- Dr Hashir further briefed the house that students are actively encouraged to engage in national and international competitions such as the Microsoft Imagine Cup, Huawei ICT Innovation Competition, Google's coding competitions, and the Cisco Global Problem Solver Challenge, among others. Moreover, the SE department's Final Year Design Projects (FYDPs) have secured funding amounting to 1 Million Rs from the Pakistan Engineering Council.
- Industrial visits are organized regularly to provide students with practical exposure to real-world industrial environments. These visits offer valuable insights into industrial processes, technologies, and best practices, enhancing the students' understanding and complementing their academic learning.
- Dr Hashir added that oral presentations have been made mandatory across all courses to enhance students' communication skills. Additionally, mock interviews are organized for final-year students to refine their interviewing skills and better prepare them for professional opportunities.

- Students are encouraged to pursue internships lasting at least 6 to 8 weeks upon completing their second and third years of their degree program.
- Approximately 85% of BSCS graduates have successfully secured opportunities in the industry, while 13% have chosen to pursue higher education.
- Based on industry recommendations from the last IAB, the curriculum has been updated to reflect current trends. A new course on Virtual and Augmented Reality and Large Language Models has been introduced. To address industry feedback regarding core programming skills, the Fundamentals of Programming course has been extended from 3 credit hours to 4 credit hours. The industry also emphasized the importance of product-based projects. Students are now actively encouraged to collaborate with industry for their projects. However, further efforts are needed to increase industry collaboration for Final Year Design Projects (FYDPs). In response to industry suggestions, students are now encouraged to file patents for their FYDPs. The NUST IP office offers comprehensive support in this regard.
- Dr. Hashir solicited input from industry stakeholders for various aspects of curriculum enhancement and engagement. These include:
    - **Suggesting topics/content:** Industry professionals are invited to propose topics or content additions to the curriculum to ensure alignment with current industry needs and trends.
    - **Delivering guest lectures:** Industry experts are encouraged to deliver guest lectures for selected courses, sharing their practical insights and experiences with students.
    - **Incorporating complex engineering problems:** Advice is sought from the industry on how to effectively incorporate complex engineering problems into courses, enriching the learning experience.
    - **Offering internships and training:** Industry partners are invited to provide internship opportunities and training programs for students, as well as short-term placements for faculty members to stay updated with industry practices.
    - **Engaging in FYDPs:** Industry collaboration is encouraged by engaging students in industry projects through the Final Year Design Projects (FYDPs), fostering practical learning experiences.
    - **Improving lab components:** Industry advice is sought to enhance the lab components of courses, ensuring they align with industry standards and practices.
    - **Providing industrial training and short courses:** Industry partners are encouraged to offer industrial training and short courses relevant to industry needs, supplementing students' education with practical skills and knowledge.

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- Dr Faisal invited participants to share their thoughts. Dr Saad added that technology is changing so fast and you can't change the basic concepts. Basic courses shouldn't be dropped. Engineering knowledge and problem-solving skills should be in all students. He further added that good-quality projects were displayed in this open house. He emphasised that the internship mechanism should be improved along with faculty liaison with the industry.
- Dr. Abdullah emphasized the importance of enhancing students' soft skills and advocated for training initiatives focused on instilling ethical practices aligned with our cultural values. Teamwork abilities can be improved among the graduates.
- Dr Hammad pointed out that there are many gaps in our education system so we should teach our students basic knowledge and math concepts of our students should be improved.
- Dr. Hammad highlighted numerous gaps within our education system, emphasizing the necessity of providing students with foundational knowledge and improving their understanding of basic mathematical concepts.
- Ms. Samina emphasized the importance of enhancing students' analytical skills to better prepare them for academic and professional success.
- Dr Faisal appreciated the feedback from the industry and mentioned that due efforts would be taken by the department to implement the suggested points.

FoC Department Coordinator
May 06, 2024

Mr Taufique ur Rehman
DLQEC SEECS
May 06, 2024